

Pokémon and Their Strengths by Type, Stats, and Generation

INFO 526 - Summer 2026 - Final Project

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Motivation and Data

Why Pokémon?

- Grew up with the franchise
- Intuitions about type
- Assumptions about strength and size
- Is any of this true?



The Dataset

- 949 Pokémon species, 22 variables
- Source: TidyTuesday (Apr 2025)
- Covers Generations 1–7
- Legendary/mythical flags joined from PokeAPI separately

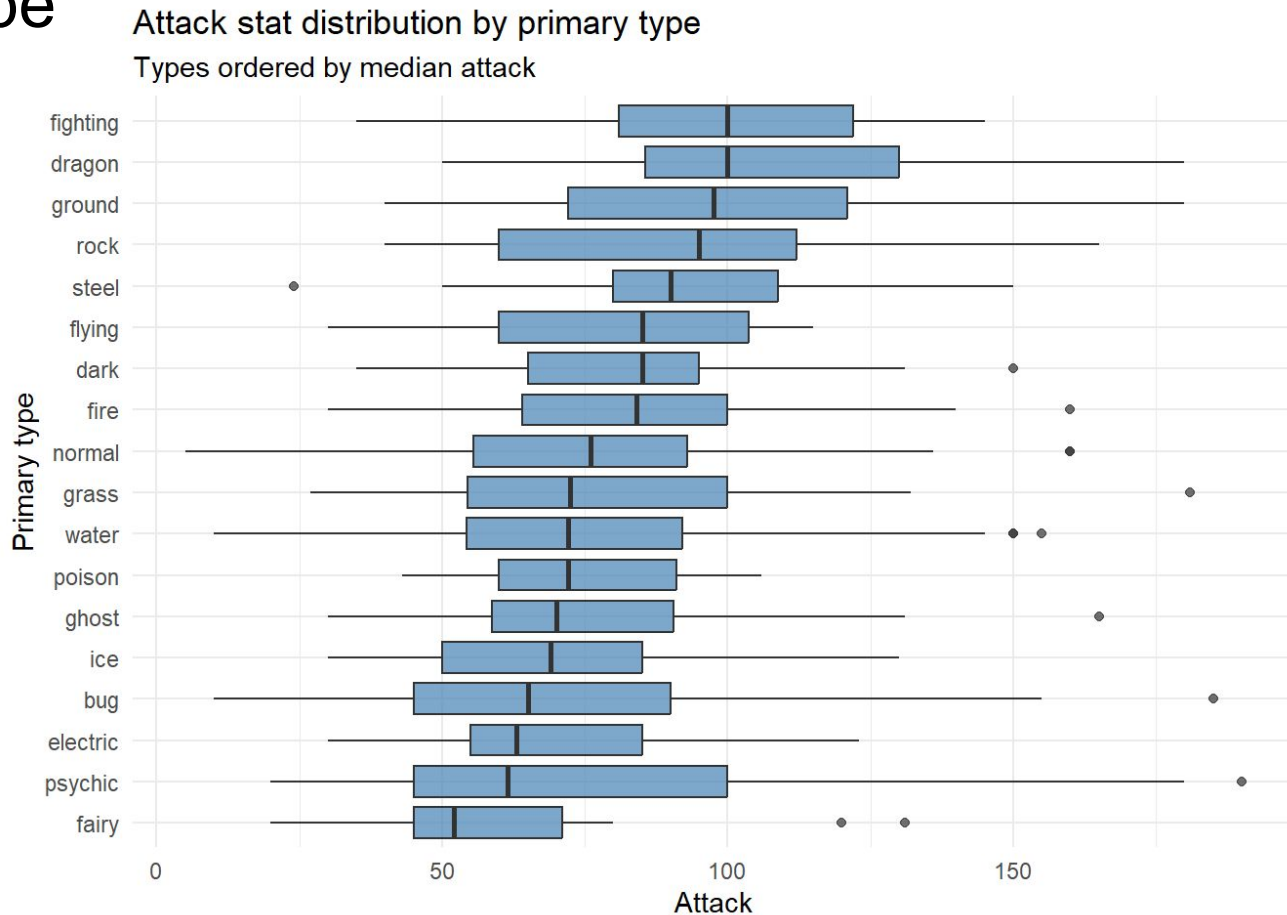


Two Questions

1. **Type, attack & weight:** Is there a relationship between a Pokémon's primary type, its attack stat, and its weight? Do types like Fighting or Dragon hit harder?
2. **Legendary status & power creep:** Do legendary and mythical Pokémon consistently outperform standard species across every generation? Is there evidence that newer generations introduced stronger Pokémon?

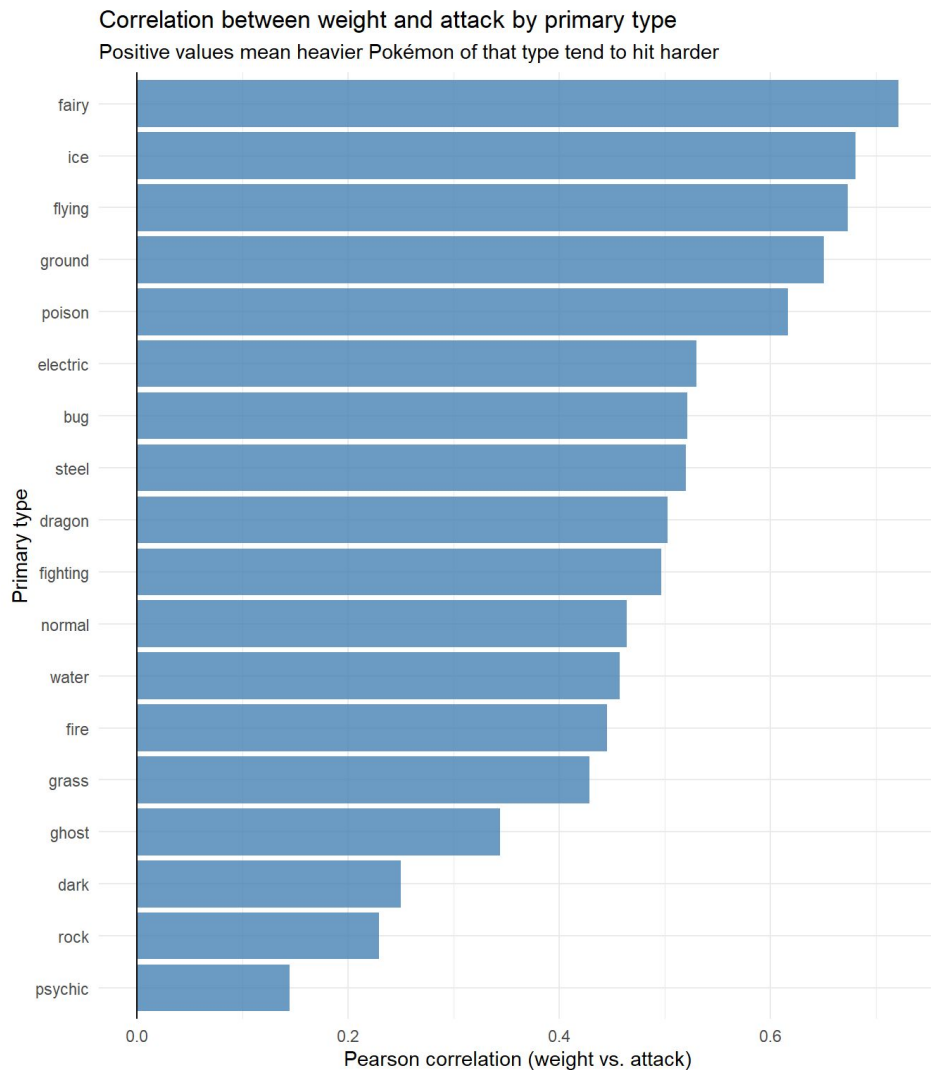
Q1 - Attack by Type

- Fighting, dragon, ground, rock are all highest on average
- Psychic, fairy are low, but have high outliers



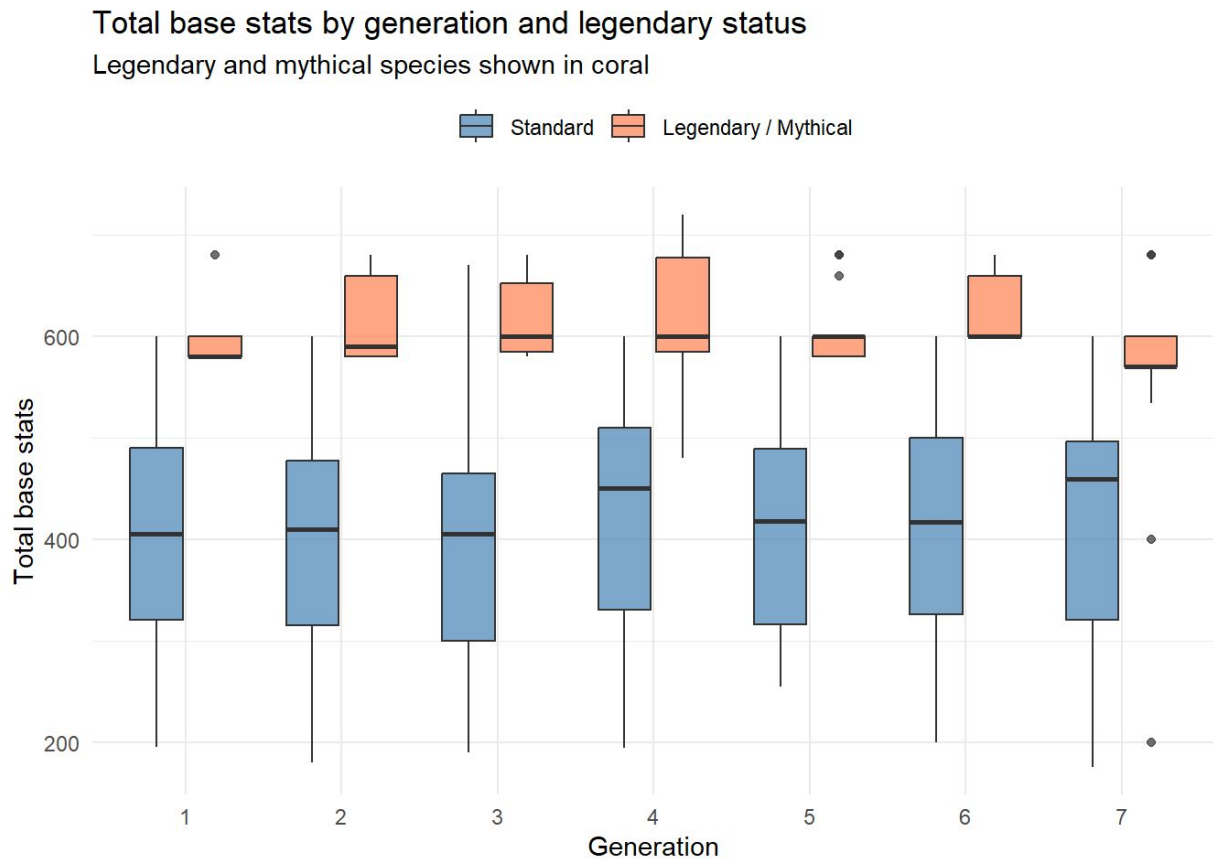
Q1 - Attack by Weight

- All are positively correlated
- Fairy, ice, flying, ground are all highest and near each other
- Psychic, rock, and dark are lowest



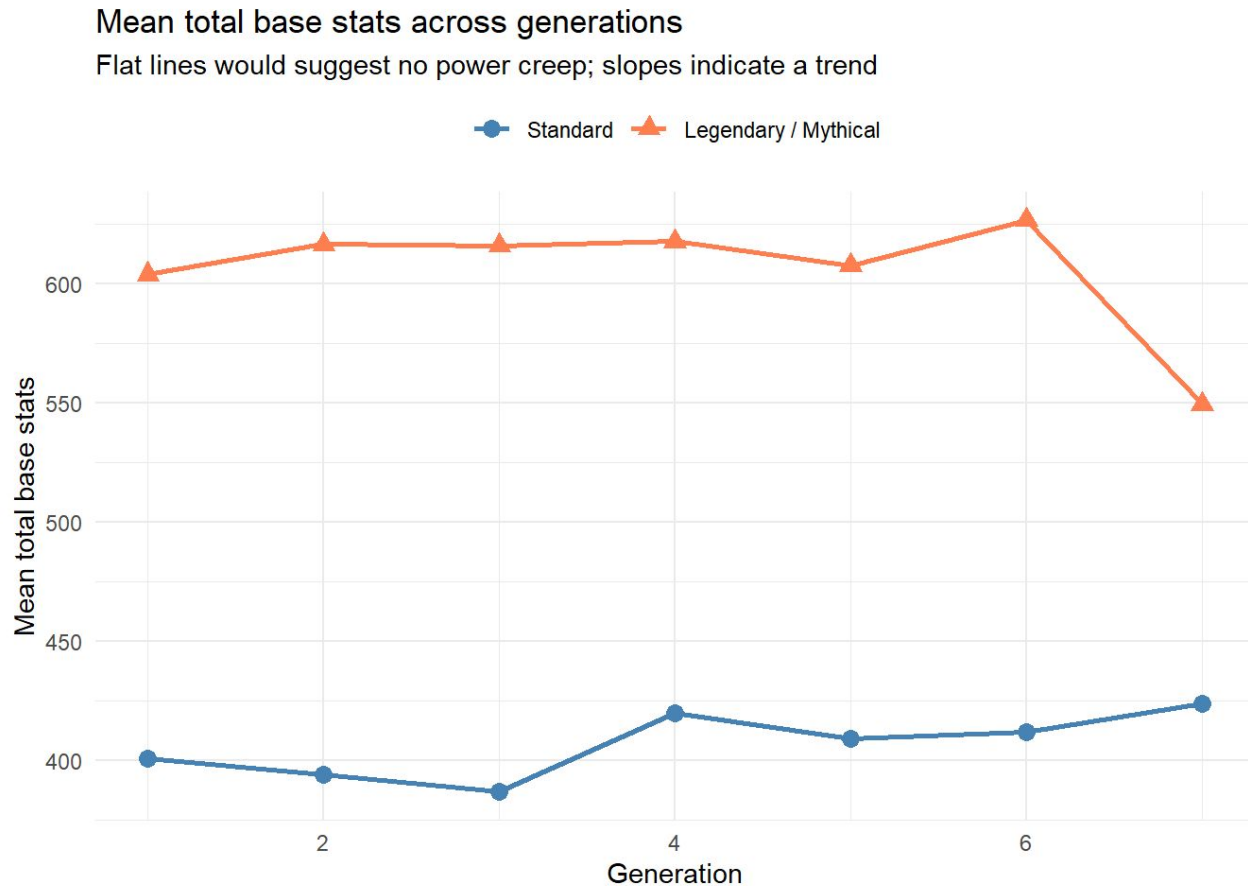
Q2 - Legendaries vs. Standard

- Legendaries are higher in all generations
- Legendary is also more tightly packed



Q2 - Power Creep

- No true power creep
- Small rise and drop from generation 6 and 7 for legendaries
- Stable



Conclusions

Q1: Yes, there is a correlation between attack, weight, and type, with Fighting, Dragon, Ground and Rock being the strongest on average by attack stat, and Fairy, Ice, Flying, and Ground, having the largest correlation between attack and weight.

Q2: Yes, legendary Pokémon consistently have higher total base stats, and there is not a strong indication of power creep over time. These stats have stayed relatively stable over generations.